

Name: _____

Date: _____

Year 11 Advanced Mathematics — Calculus Review

Limits, Differentiation and Integration

Learning Intentions

By the end of this worksheet you will be able to:

- Evaluate limits using algebraic techniques
- Differentiate polynomial, power, and composite functions
- Apply integration as the reverse of differentiation
- Identify and correct common calculus errors

Section A: Fluency

1. Evaluate each limit. Show any algebraic manipulation required.

- (a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
- (b) $\lim_{x \rightarrow 0} \frac{5x^3 - 2x}{x}$
- (c) $\lim_{x \rightarrow \infty} \frac{4x^2 + 1}{2x^2 - 3}$

2. Differentiate each function with respect to x . Simplify your answer fully.

- (a) $f(x) = 3x^4 - 7x^2 + 2x - 5$
- (b) $g(x) = \frac{4}{x^3} + \sqrt{x}$
- (c) $h(x) = (2x - 1)^5$

Show working clearly

Page 1

AI Draft | Teacher to review

5. A student evaluates $\int_1^3 (2x + 1) \, dx$ as follows:

Student's Working:

$$\begin{aligned} \int_1^3 (2x + 1) \, dx &= [x^2 + x + C]_1^3 \\ &= (9 + 3 + C) - (1 + 1 + C) \\ &= 12 + C - 2 - C \\ &= 10 \end{aligned}$$

The answer is $10 + C$.

- (a) The student's final numerical value of 10 is correct, but they have made a conceptual error in their final statement. Identify and explain it.
- (b) State the correct final answer and explain why C does not appear in a definite integral result.

Section C: Investigation Skills

Section C Instructions

These questions ask you to explore relationships, identify variables, and draw evidence-based conclusions. Show all reasoning clearly.

6. Investigation: Gradient of a Curve

Consider the function $f(x) = x^3 - 3x$.

- (a) Find $f'(x)$.
- (b) Complete the table below by evaluating $f'(x)$ at each value of x .

x	-2	-1	0	1	2
$f'(x)$					

Show working clearly

Page 3

AI Draft | Teacher to review

3. Find each indefinite integral. Include the constant of integration.

- (a) $\int (6x^2 - 4x + 3) \, dx$
- (b) $\int x^{-4} \, dx$
- (c) $\int (3x + 1)^2 \, dx$

Section B: Spot the Error

Section B Instructions

Each question below contains exactly **one error** in the student's working. For each question:

- Identify the **line** where the error occurs.
- State **what the error is** and why it is incorrect.
- Provide the **correct solution**.

4. A student differentiates $f(x) = (x^2 + 3)^4$ as follows:

Student's Working:

$$\begin{aligned} f(x) &= (x^2 + 3)^4 \\ f'(x) &= 4(x^2 + 3)^3 \end{aligned}$$

- (a) On which line does the error occur? What rule has the student forgotten to apply?
- (b) Write the correct derivative, showing full working.

Show working clearly

Page 2

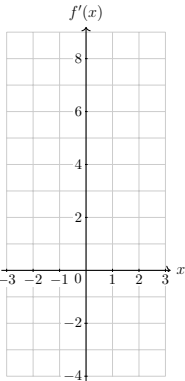
AI Draft | Teacher to review

(c) Using your table, identify the x -values where the gradient is zero. What do these points represent on the curve?

(d) **Independent variable (IV):** What are you changing in this investigation?

Dependent variable (DV): What are you measuring?

(e) Sketch $f'(x)$ on the axes below. Label the x -intercepts.



(f) Write one evidence-based conclusion linking the sign of $f'(x)$ to the behaviour (increasing or decreasing) of $f(x)$.

Show working clearly

Page 4

AI Draft | Teacher to review

Name: _____

Date: _____

Year 11 Advanced Mathematics — Calculus Review

Learning intentions (teacher-review draft)

- Review differentiation of polynomials, simple products, and quotients.
- Interpret derivatives graphically and in context.
- Identify and correct common calculus misconceptions.
- Apply investigation skills: variables, data, reasoning, and conclusions.

1. Differentiate each function. Show all working.

(a) $f(x) = 3x^4 - 5x^2 + 7$

(b) $g(x) = \frac{2}{x^3}$

(c) $h(x) = (x^2 + 1)(3x - 4)$

Show working clearly

Page 1

AI Draft | Teacher to review

Name: _____

Date: _____

Year 11 Mathematics Advanced – Calculus Review

Teacher-review draft aligned to Stage 6 introductory differential calculus.

Learning Intentions

- Differentiate polynomial and simple power functions.
- Interpret gradients and rates of change.
- Identify and correct common differentiation errors.
- Investigate how changing parameters affects gradients.

1. Differentiate the following functions.

(a) $f(x) = 3x^4 - 2x^2 + 7$

(b) $g(x) = 5x^3 + 4x - 9$

(c) $h(x) = \frac{1}{2}x^5 - 3x^2$

2. For $y = x^3 - 6x$:

(a) Find $\frac{dy}{dx}$.

(b) Determine the gradient when $x = 2$.

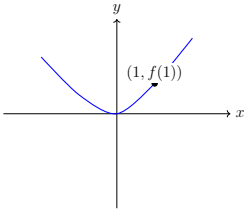
(c) State whether the function is increasing or decreasing at $x = 2$.

Show working clearly

Page 1

AI Draft | Teacher to review

2. The graph below shows a sketch of $y = f(x)$. Estimate $f'(x)$ at $x = 1$ and explain your reasoning.



3. **Spot the Error.** A student differentiates $k(x) = \frac{x^2 - 3}{x}$ as follows:

$$k'(x) = \frac{2x}{x} - \frac{x^2 - 3}{x^2}.$$

(a) Identify the exact step where the error occurs.

(b) Correct the working.

(c) Explain why the student's method fails in general.

Show working clearly

Page 2

AI Draft | Teacher to review

3. **Spot the Error**

A student differentiates $y = 4x^3 - 7x^2 + 2$ as:

$$\frac{dy}{dx} = 12x^3 - 14x.$$

(a) Identify the exact error.

(b) Write the correct derivative.

(c) Explain why the student's method is incorrect.

4. **Spot the Error**

A student claims:

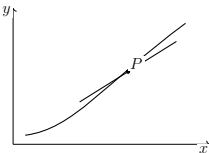
$$\frac{d}{dx}(8) = 8.$$

(a) Explain the misconception.

(b) State the correct derivative.

(c) Give another example involving a constant.

5. The graph below represents a function at a point P .



Show working clearly

Page 2

AI Draft | Teacher to review

Name: _____

Date: _____

Year 11 Mathematics Advanced: Calculus Review
Spot the Error + Investigation Skills

Teacher-review draft

This worksheet reviews introductory calculus ideas for Stage 6 Mathematics Advanced. It focuses on gradient, average rate of change, instantaneous rate of change, differentiation, tangent lines, and interpretation from data.

Learning intentions

By the end of this worksheet, you should be able to:

- connect gradient, rate of change, and the derivative;
- differentiate simple polynomial functions;
- find tangent and normal equations;
- identify and correct common calculus errors;
- use data to estimate and interpret rates of change.

Key idea

Average rate of change over $[a, b]$ is $\frac{f(b) - f(a)}{b - a}$.

Instantaneous rate of change at $x = a$ is represented by $f'(a)$.

1. Access starter: average rate of change

A function is given by

$$f(x) = x^2 + 2x.$$

- (a) Find $f(1)$ and $f(4)$.
- (b) Find the average rate of change of $f(x)$ from $x = 1$ to $x = 4$.
- (c) Explain what the average rate of change represents on a graph.

Show working clearly

Page 1

AI Draft | Teacher to review

4. Fluency: differentiate

Differentiate each function.

- (a) $y = 5x^4 - 3x^2 + 7$
- (b) $y = \frac{6}{x^2}$
- (c) $y = (2x - 1)(x + 4)$
- (d) $y = 3\sqrt{x}$

5. Spot the Error: power rule

A student differentiates

$$y = \frac{8}{x^3}$$

as follows:

$$y = 8x^3$$

$$\frac{dy}{dx} = 24x^2.$$

- (a) Identify the exact error.
- (b) Correct the working.
- (c) Explain the key rule the student forgot.

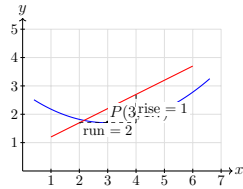
Show working clearly

Page 3

AI Draft | Teacher to review

2. Gradient from a graph

The diagram below shows a curve and a tangent line at point P .



- (a) Use the tangent triangle to estimate the gradient at P .
- (b) State the approximate value of $f'(3)$.
- (c) Explain why the tangent gradient is more useful than a secant gradient when describing instantaneous rate of change.

3. First principles

Use first principles to find the derivative of

$$f(x) = x^2 + 3x.$$

Start from

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Show working clearly

Page 2

AI Draft | Teacher to review

6. Tangent and normal

The curve is

$$y = x^2 - 4x + 1.$$

- (a) Find $\frac{dy}{dx}$.
- (b) Find the gradient of the tangent at $x = 3$.
- (c) Find the coordinates of the point on the curve when $x = 3$.
- (d) Find the equation of the tangent at $x = 3$.
- (e) Find the equation of the normal at $x = 3$.

7. Investigation skills: motion from data

A ball rolls along a straight track. Its distance from the starting point is recorded below.

Time, t (s)	Distance, s (m)
0	0.0
1	0.7
2	2.8
3	6.3
4	11.2
5	17.5

- (a) Calculate the average velocity from $t = 1$ to $t = 4$.
- (b) Estimate the instantaneous velocity at $t = 3$ using the symmetric interval from $t = 2$ to $t = 4$.
- (c) State one controlled condition that should be kept constant during the experiment.
- (d) State one possible source of measurement error.
- (e) The data appears to follow $s = 0.7t^2$. Use calculus to find the velocity function.

Show working clearly

Page 4

AI Draft | Teacher to review

Name: _____

Date: _____

Year 11 Mathematics Advanced
Introduction to Calculus: Stationary Points Investigation

Syllabus Focus: MA-C1 Introduction to Calculus

This worksheet reviews foundational differential calculus concepts, focusing on the formal definition of the derivative (first principles), the geometric interpretation of gradients, and algebraic differentiation rules.

1. Spot the Error (First Principles)

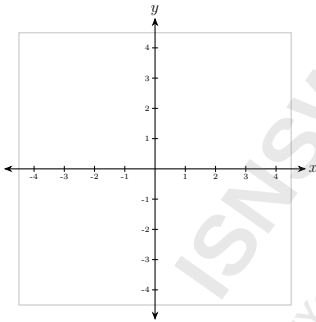
A student is asked to differentiate $f(x) = 3x^2 - 5x$ from first principles. Their working is shown below:

$$\begin{aligned} f'(x) &= \frac{f(x+h) - f(x)}{h} \\ &= \frac{3(x+h)^2 - 5x + h - (3x^2 - 5x)}{h} \quad \text{--- Step 1} \\ &= \frac{3(x^2 + 2xh + h^2) - 5x + h - 3x^2 + 5x}{h} \quad \text{--- Step 2} \\ &= \frac{3x^2 + 6xh + 3h^2 + h - 3x^2}{h} \quad \text{--- Step 3} \\ &= \frac{6xh + 3h^2 + h}{h} \quad \text{--- Step 4} \\ &= 6x + 3h + 1 \quad \text{--- Step 5} \\ &= 6x + 1 \quad \text{--- Step 6} \end{aligned}$$

Identify the **two** distinct algebraic or conceptual errors made by the student. Explain why each is incorrect and provide the completely corrected full working.

Show working clearly

Page 1



Show working clearly

Page 3

2. Geometric Investigation Skills: Curve Analysis & Stationary Points

Consider the curve defined by the function $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$.

- a. Determine the algebraic expression for the gradient function, $f'(x)$.
- b. Find the coordinates of the points on the curve where the tangent is completely horizontal (parallel to the x -axis).
- c. Construct a first-derivative sign table for the critical values found in part (b) to classify the nature of these stationary points.

d. State the intervals of x for which the curve $y = f(x)$ is strictly increasing.

e. Investigate the behavior of the gradient across the domain by sketching the gradient function $y = f'(x)$ on the grid below. Clearly label all intercepts.

Show working clearly

Page 2

3. Algebraic Mastery & Parameter Shift

Differentiate the following functions with respect to x . Express your answers with positive indices and in radical form where applicable.

- a. $f(x) = \frac{4}{x^3} - \sqrt{x}$
- b. Given that $g(x) = ax^n + bx^2$, where a and b are constants, it is known that $g'(1) = 10$ and $g'(2) = 32$. Determine the exact values of parameters a and n if n is a positive integer greater than 2.

Show working clearly

Page 4

Name: _____

Date: _____

Year 11 Calculus Review: Derivatives

Key Idea: The derivative measures the rate of change of a function. Use rules such as power, product, and chain rules.

1. Find the derivative of the following functions:

- (a) $f(x) = 3x^4 - 5x^2 + 7$
- (b) $g(x) = \frac{1}{x^2}$
- (c) $h(x) = (2x + 3)^5$

2. Find the gradient of the tangent to $y = x^3 - 4x$ at $x = 2$.

3. A particle moves along a line with position given by $s(t) = 2t^3 - 3t^2 + t$.

- (a) Find the velocity function.
- (b) Find the acceleration function.
- (c) Determine when the particle is at rest.

4. Spot the Error

A student differentiates $y = (x^2 + 1)^3$ as follows:

$$\frac{dy}{dx} = 3(x^2 + 1)^2$$

- (a) Identify the mistake.
- (b) Correct the derivative.
- (c) Explain why the chain rule is required.

5. Spot the Error

A student claims:

$$\frac{d}{dx}(x \sin x) = \sin x$$

- (a) Identify the mistake.
- (b) State the correct derivative.
- (c) Name the rule used.

6. Investigation Skills

Consider the function $f(x) = x^2$.

(a) Complete the table of average rates of change between $x = 1$ and nearby points:

x	Average rate of change
1.5	
1.1	
1.01	

- (b) What value do these rates approach?
- (c) Form a hypothesis for the gradient at $x = 1$.

Name: _____

Date: _____

Calculus: Spot the Error & Kinematic Investigation

Learning Intentions:

- Identify and correct algebraic errors in First Principles differentiation.
- Apply the power rule fluently to terms with negative and fractional indices.
- Investigate instantaneous rates of change in a physical context (kinematics).

1. Fluency: The Power Rule

Differentiate each of the following functions with respect to x . Express your final answers with positive indices where applicable.

(a) $y = 3x^4 - 5x^2 + 7x - 2$

(b) $f(x) = \frac{4}{x^3} - \sqrt{x}$

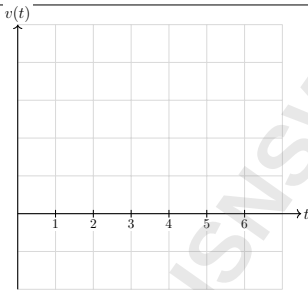
2. Diagnostic: Spot the Error

A student was asked to find the derivative of $f(x) = x^2 - 4x$ using differentiation from first principles. Their working is shown below:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4x - (x^2 - 4x)}{h} \quad \dots\dots \text{(Line 1)} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + h^2 - 4x - x^2 + 4x}{h} \quad \dots\dots \text{(Line 2)} \\ &= \lim_{h \rightarrow 0} \frac{h^2}{h} \\ &= \lim_{h \rightarrow 0} h = 0 \end{aligned}$$

Show working clearly

Page 1



4. Synthesis: Parameter Shift

The curve $y = ax^2 + bx$ has a tangent with a gradient of 4 at the point (1, 3). By investigating the properties of the gradient function and the given coordinates, set up simultaneous equations to find the exact values of the constants a and b .

Show working clearly

Page 3

(a) Identify and explain the **two** distinct mathematical errors the student made in Lines 1 and 2.

(b) Provide the correct full working to find $f'(x)$ from first principles.

3. Investigation Skills: Motorcycle Kinematics

A motorcycle accelerates away from a set of traffic lights along a straight road. Its displacement, s metres, from the traffic lights after t seconds is modelled by the function:

$s(t) = 2t^3 - 15t^2 + 36t \quad \text{for } t \geq 0$

(a) Determine the velocity function, $v(t)$, of the motorcycle.

(b) Investigate the times at which the motorcycle is instantaneously at rest.

(c) Construct a sign table to investigate the intervals where the velocity is positive (moving forwards) and negative (moving backwards).

(d) On the grid below, sketch the velocity-time graph to support your investigation findings from parts (b) and (c).

Show working clearly

Page 2

Teacher Answer Key & Marking Guide

1. Fluency: The Power Rule

(a) $y' = 12x^3 - 10x + 7$ [1 mark for each correct term]

(b) $f(x) = 4x^{-3} - x^{1/2}$ (Index form setup)
 $f'(x) = -12x^{-4} - \frac{1}{2}x^{-1/2} = -\frac{12}{x^4} - \frac{1}{2\sqrt{x}}$ [1 mark for negative index rule, 1 mark for fractional index rule]

2. Diagnostic: Spot the Error

(a) **Error 1 (Line 1):** The student failed to substitute $(x + h)$ into the $-4x$ term. It should read $-4(x + h)$, not just $-4x$.
Error 2 (Line 2): The student incorrectly expanded the perfect square $(x + h)^2$ as $x^2 + h^2$. They missed the middle term $2xh$. [1 mark for each correctly identified error]

(b) Correct working:
$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 4(x+h) - (x^2 - 4x)}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - x^2 + 4x}{h}$$
$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h} = \lim_{h \rightarrow 0} (2x + h - 4) = 2x - 4 \quad [2 \text{ marks for correct expansion and limit evaluation}]$$

3. Investigation Skills: Motorcycle Kinematics

(a) $v(t) = s'(t) = 6t^2 - 30t + 36$ [1 mark]

(b) Set $v(t) = 0$:
 $6(t^2 - 5t + 6) = 0 \implies 6(t - 2)(t - 3) = 0$
The motorcycle is at rest at $t = 2$ and $t = 3$ seconds. [1 mark factorisation, 1 mark solutions]

(c) Sign table:
 $t = 1 \implies v(1) = 6(1)^2 - 30(1) + 36 = 12$ (+)
 $t = 2.5 \implies v(2.5) = 6(-0.5)(0.5) = -1.5$ (-)
 $t = 4 \implies v(4) = 6(2)(1) = 12$ (+)
Moves forwards for $0 \leq t < 2$ and $t > 3$. Moves backwards for $2 < t < 3$. [2 marks for valid table/test points]

(d) Graph should be a concave-up parabola. t -intercepts at $t = 2$ and $t = 3$. v -intercept at $v = 36$ (will be off the provided standard scale, look for correct shape and roots). [2 marks for roots and correct concavity]

4. Synthesis: Parameter Shift

Show working clearly

Page 4

Universal Worksheet Engine Showcase

A pdfLaTeX-safe portfolio of printable classroom artefacts

This portfolio demonstrates one engine producing structured worksheets across primary learning, STEM, HSC-style mathematics, science practicals, humanities analysis, computing, diagrams, graphs, circuits, chemical notation, answer keys, and QR-supported teacher workflows.

**Teacher draft
review before use**

Design principle: notation, diagrams, worked space, and marking guidance are not decoration; they are part of the pedagogy.

Capability Map

- Primary visual learning
- Measurement and ruler work
- Geometry and transformations
- Statistics, probability, networks
- HSC calculus and vectors
- Science practical design
- Physics, chemistry, biology
- Humanities, English, business
- Computing, logic and algorithms
- Teacher answer key and QR support

Safe Engineering Choices

- One pdfLaTeX document
- No external images
- No fontspec
- Sans-serif via `\sfdefault`
- TikZ/PGF graphics only
- White-backed labels
- Bounded graphs and diagrams
- Footer reinforces review

Pedagogical Coverage

- Access starters
- Representation shifts
- Misconception checks
- Parameter shifts
- Practical investigation skills
- Exam-style prompts
- Proof and generalisation
- Multi-step synthesis

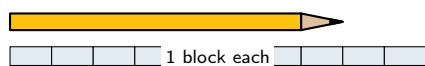
Name: _____

Date: _____

Part A: Universal Cross-Curricular Sampler

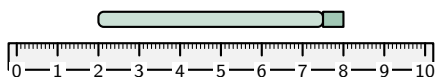
Dense layouts, diagrams, graphs, symbolic notation, practical skills and answer spaces in one coherent printable system.

1. **Primary Measurement:** Count the blocks. How long is the pencil? [1 mark]

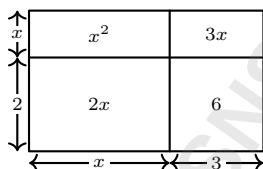


The pencil is _____ blocks long.

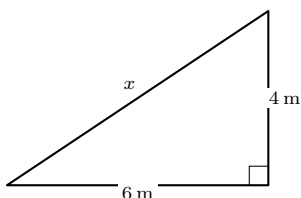
2. **Spot the Measuring Error:** A student says the marker is 5 cm long. Explain the error and give the correct length. [2 marks]



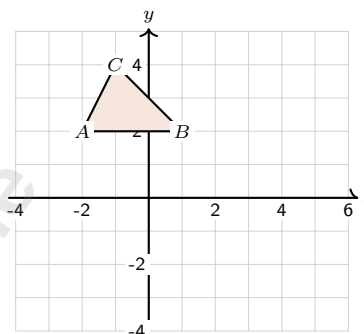
3. **Algebra Representation:** The rectangle shows $(x + 3)(x + 2)$. Expand the expression and label each region. [2 marks]



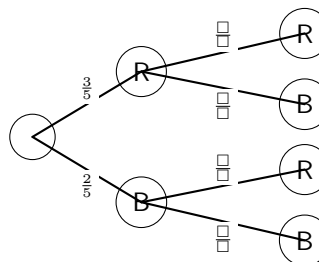
4. **Geometry:** Find x to one decimal place. [2 marks]



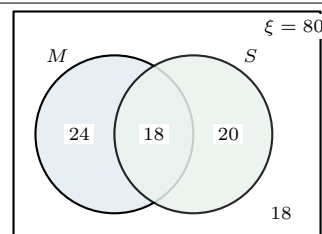
5. **Transformations:** Triangle ABC is translated by vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Draw the image and write A' , B' , C' . [3 marks]



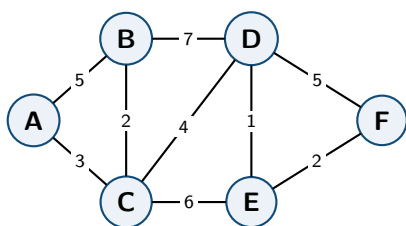
6. **Probability Tree:** A bag has 3 red and 2 blue counters. Two counters are drawn without replacement. Complete the tree and calculate $P(\text{two red})$. [3 marks]



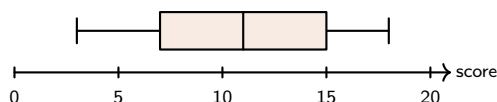
7. **Set Theory:** Use the Venn diagram to find $n(M \cap S)$ and $P(S | M)$. [3 marks]



- 8. Networks:** Determine the shortest path from *A* to *F* and its total weight. **[3 marks]**



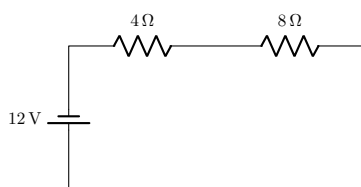
- 9. Statistics:** The box plot summarises quiz scores out of 20. State the median and interquartile range. **[2 marks]**



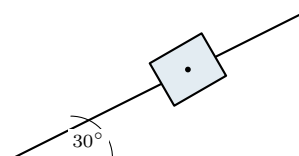
- 10. Science Investigation Skills:** A student tests whether salt affects plant growth. They use different seeds, different pots, and different window positions. Identify the independent variable, dependent variable, and two controlled variables that should be kept the same. **[4 marks]**

Variable type	Response
Independent variable	
Dependent variable	
Controlled variable 1	
Controlled variable 2	

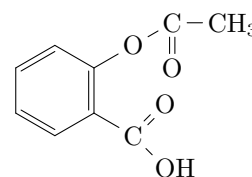
- 11. Physics - Circuits:** For the circuit below, calculate the total resistance and current if $V = 12\text{ V}$. **[3 marks]**



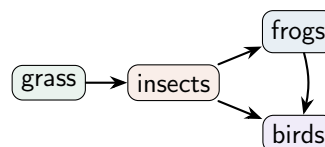
- 12. Physics - Free Body Diagram:** Draw and label the weight, normal reaction, and friction force acting on the block. **[3 marks]**



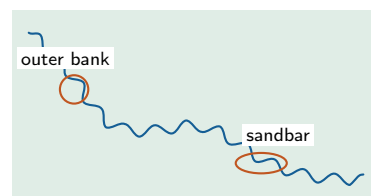
- 13. Chemistry:** Identify two functional groups in aspirin and write the equation for salicylic acid reacting with acetic anhydride to form aspirin and ethanoic acid. **[4 marks]**



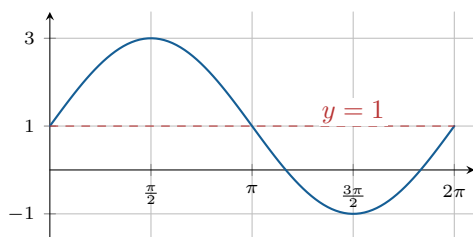
- 14. Biology - Food Web:** If the insect population suddenly falls, explain the likely short-term effect on frogs and birds. **[3 marks]**



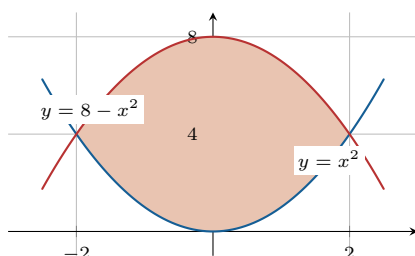
- 15. Earth Science:** Label one example each of erosion, transport, and deposition on the river diagram. **[3 marks]**



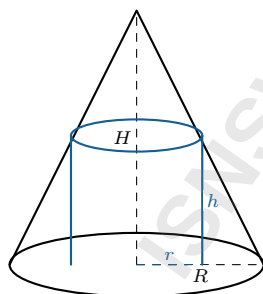
- 16. Trigonometry Graphing:** For $y = 2\sin(x) + 1$, state the amplitude, midline, and range. [3 marks]



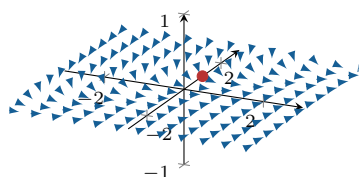
- 17. Calculus - Area:** Find the exact area bounded by $y = x^2$ and $y = 8 - x^2$. [4 marks]



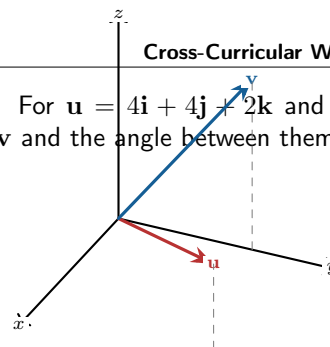
- 18. Calculus - Parameter Shift:** A cylinder is inscribed in a cone. Explain how increasing R while holding H fixed affects the maximum volume. [3 marks]



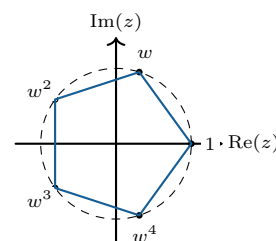
- 19. Differential Equations:** Sketch the solution curve through $(0, 1)$ for $\frac{dy}{dx} = x - y$. [3 marks]



- 20. 3D Vectors:** For $\mathbf{u} = 4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 3\mathbf{j} + 4\mathbf{k}$, calculate $\mathbf{u} \cdot \mathbf{v}$ and the angle between them. [4 marks]



- 21. Complex Numbers:** The diagram shows the fifth roots of unity. Prove $1 + w + w^2 + w^3 + w^4 = 0$. [3 marks]

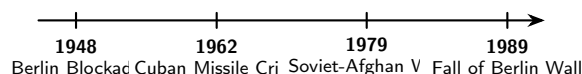


- 22. Business Studies - Critical Reasoning:** A company has already spent \$500 million on an old engine design. Electric vehicles then become cheaper to produce. The board says, "We must launch it because we already paid for it." Name the fallacy and explain the better decision rule. [3 marks]

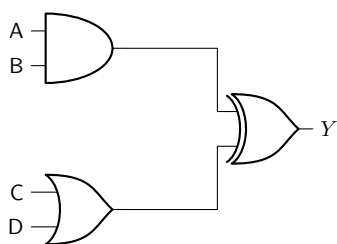
- 23. English - Literary Analysis:** Identify the main technique in the phrase "a sea of troubles" and explain its effect. [3 marks]

Whether 'tis nobler in the mind to suffer the slings and arrows of outrageous fortune, or to take arms against a sea of troubles...

- 24. History - Timeline Analysis:** Choose one event and explain one long-term consequence. [3 marks]



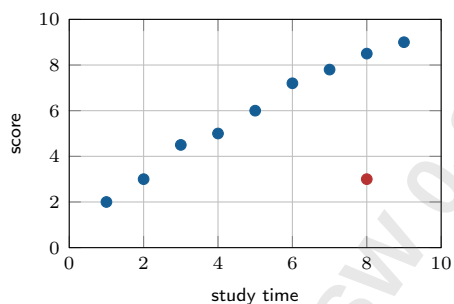
- 25. Computing - Digital Logic:** Write the Boolean expression for Y . [2 marks]



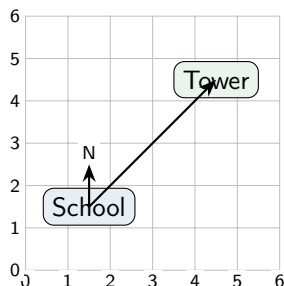
- 26. Algorithm Trace:** The recurrence is $a_1 = 3$ and $a_n = 2a_{n-1} - 1$. Complete the trace table to find a_4 . [2 marks]

n	1	2	3	4
a_n	3			

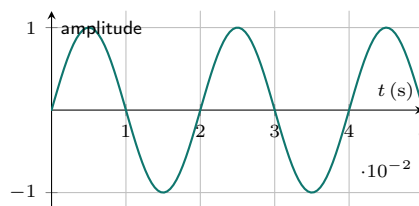
- 27. Data and Modelling:** Describe the association in the scatterplot and identify one possible outlier. [3 marks]



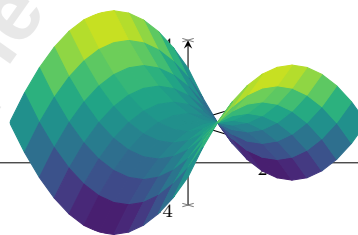
- 28. Geography - Grid Reference:** Give the grid reference for the water tower and estimate the bearing from the school to the tower. [3 marks]



- 29. Music/Physics - Waves:** The sound wave below has a period of 0.02 s. Calculate its frequency. [2 marks]



- 30. Extension - Multivariable Visualisation:** The surface is $z = x^2 - y^2$. Mark whether the point $(1, 1, 0)$ lies on the surface and explain why. [2 marks]



Part B: Year 2 Measurement Demo

Primary-level visual engagement using pure TikZ: words, objects, informal units, rulers and hands-on measuring.

Learning Intention

I can compare and measure length using informal units and centimetres. I can explain why starting at zero matters when using a ruler.

1. **Match the words.** Draw a line from each word to the matching picture.

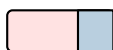
String



Rubber



Tall

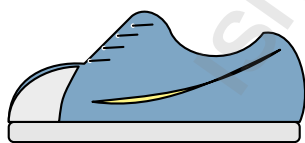


2. **Word bank.** Use the words to finish the sentence.

string tall rubbers

To find out how _____ my tower is, I can measure it with _____.

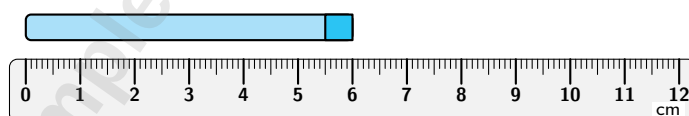
3. **How many rubbers long is the shoe?**



RUBBER RUBBER RUBBER RUBBER RUBBER

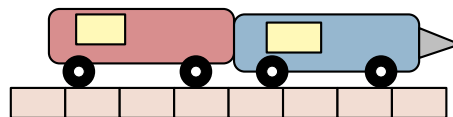
The shoe is _____ rubbers long.

4. **Use the ruler.** Measure the marker in centimetres.



The marker is _____ cm long.

5. **Toy train challenge.** Count the cubes under the train.

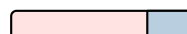


The train is _____ cubes long.

6. **Which is longer?** Circle the longer object.



pencil



rubber

Teacher Answer Key and Marking Notes

- Q1.** 8 blocks. Award 1 for correct count from start to tip.
- Q2.** Error: the object starts at 2 cm, not zero. Correct length: $8 - 2 = 6$ cm if reading the cap end at 8.
- Q3.** $(x+3)(x+2) = x^2 + 5x + 6$. Regions: x^2 , $3x$, $2x$, 6 .
- Q4.** $x = \sqrt{6^2 + 4^2} = \sqrt{52} \approx 7.2$ m.
- Q5.** $A'(1, 0)$, $B'(4, 0)$, $C'(2, 2)$.
- Q6.** After red first: $2/4$ red, $2/4$ blue. $P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$.
- Q7.** $n(M \cap S) = 18$. $P(S | M) = \frac{18}{24+18} = \frac{3}{7}$.
- Q8.** Shortest path $A \rightarrow C \rightarrow D \rightarrow E \rightarrow F$ with total $3+4+1+2 = 10$.
- Q9.** Median = 11. IQR = $15 - 7 = 8$.
- Q10.** IV: salt amount/concentration. DV: plant growth. Controls: seed type, pot, soil, light, water, time.
- Q11.** Series resistance $R_T = 4 + 8 = 12 \Omega$. Current $I = V/R = 12/12 = 1$ A.
- Q12.** Weight vertically down, normal perpendicular to plane, friction parallel to plane opposing likely motion.
- Q13.** Functional groups: carboxylic acid and ester; aromatic ring also acceptable. Equation: $C_7H_6O_3 + C_4H_6O_3 \rightarrow C_9H_8O_4 + C_2H_4O_2$.
- Q14.** Frogs likely decrease because they have less food. Birds may decrease if they depend on insects/frogs, but the effect may lag or depend on alternative food.
- Q15.** Erosion: outer bank; transport: moving river load; deposition: sandbar/inside bend.
- Q16.** Amplitude 2, midline $y = 1$, range $[-1, 3]$.
- Q17.** Intersections $x = \pm 2$. Area = $\int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$ square units.
- Q18.** Since $V_{\max} = \frac{4}{27}\pi R^2 H$, increasing R while holding H fixed increases maximum volume quadratically.
- Q19.** Sketch should pass through $(0, 1)$ and initially slope -1 , then curve toward the solution behaviour of $y = x - 1 + 2e^{-x}$.
- Q20.** $\mathbf{u} \cdot \mathbf{v} = 0 + 12 + 8 = 20$. $|\mathbf{u}| = 6$, $|\mathbf{v}| = 5$, so $\cos \theta = 20/30$ and

$$\theta \approx 48^\circ.$$

$$\mathbf{Q21.} w^5 = 1 \text{ and } w \neq 1, \text{ so } 1 + w + w^2 + w^3 + w^4 = \frac{1-w^5}{1-w} = 0.$$

Q22. Sunk cost fallacy. Decision should depend on future marginal costs, benefits, opportunity cost, and market fit, not unrecoverable spending.

Q23. Metaphor. It turns mental struggle into an overwhelming physical image, emphasising distress and scale.

Q24. Open response. Examples: Cuban Missile Crisis intensified nuclear caution and contributed to later arms-control communication.

$$\mathbf{Q25.} Y = (A \wedge B) \oplus (C \vee D).$$

$$\mathbf{Q26.} a_2 = 5, a_3 = 9, a_4 = 17.$$

Q27. Strong positive association; possible outlier $(8, 3)$.

Q28. Tower approximately $(4.5, 4.5)$ or grid square around $E5$ depending on local convention. Bearing from school to tower is about 045° .

$$\mathbf{Q29.} f = 1/T = 1/0.02 = 50 \text{ Hz.}$$

$$\mathbf{Q30.} \text{Yes. If } x = 1 \text{ and } y = 1, \text{ then } z = 1^2 - 1^2 = 0.$$

Year 2 Q1. Teacher checks matching: String to wavy line, Rubber to eraser, Tall to vertical height arrow.

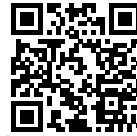
Year 2 Q2. "tall" and "rubbers".

Year 2 Q3. About 4 rubbers if aligned to the displayed informal units.

Year 2 Q4. 6 cm.

Year 2 Q5. 8 cubes.

Year 2 Q6. Pencil is longer.



Generate more



ISNSW example context